

Intermediate Macroeconomics: Lecture W8 (Ch.14)

ECON 311 – Intermediate Macroeconomics

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What are You Going to Learn (Ch. 14)

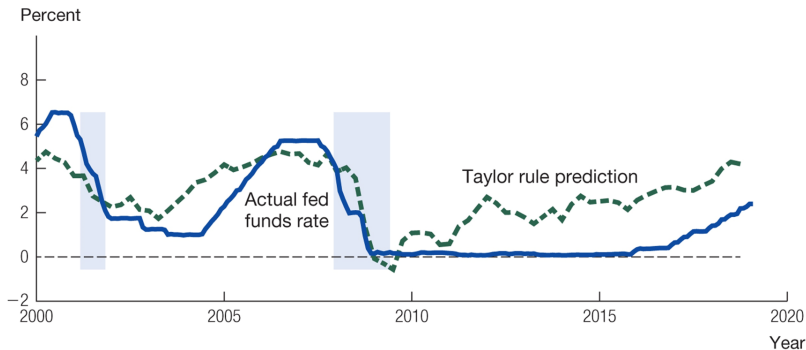
- Various policy responses to the financial crisis
- The Fed's balance sheet
- Contributors to a very large negative shock in AD
 - Gap between the fed funds rate and market interest rates
 - A household balance sheet crisis
 - Substantial uncertainty
- Unconventional actions of the Fed

Policy Responses to the Financial Crisis

- Simple policy rule: the central bank rate as a function of the gap between the current inflation rate and target rate
- The Taylor rule expands upon this: allows the current level of short-run output to influence the fed funds rate
 - e.g., $i_t = 1\% + 1.5\pi_t + 0.5\widetilde{Y}_t$

The Fed Funds Rate and the Taylor Rule

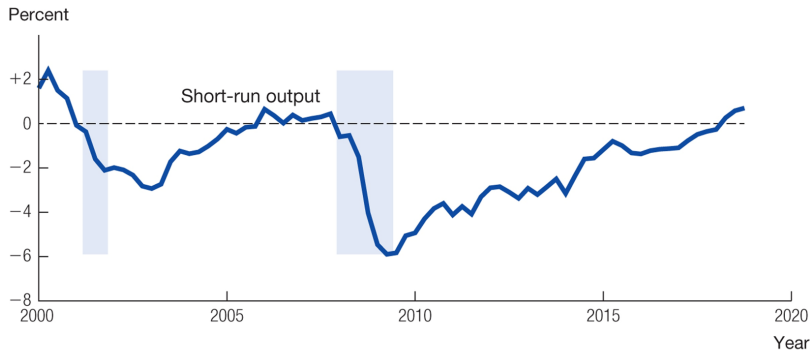
- Some people argue that leading up to the crisis, the fed funds rate may have been too low for too long
- But, during the crisis, the actual interest rate that firms faced was high
 - from 2002 to 2006 the fed funds rate was lower than the Taylor rule suggested
- After the crisis, Taylor rule suggests a higher interest rate than the Fed Rate



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The Fed Funds Rate and the Taylor Rule

- Output was below potential throughout 2002 and into 2003
 - led to inflation being well below its target, and contributed to the federal funds rate remaining low
- Short run output has since grown beyond potential, and contributed to the increase in the fed funds rate



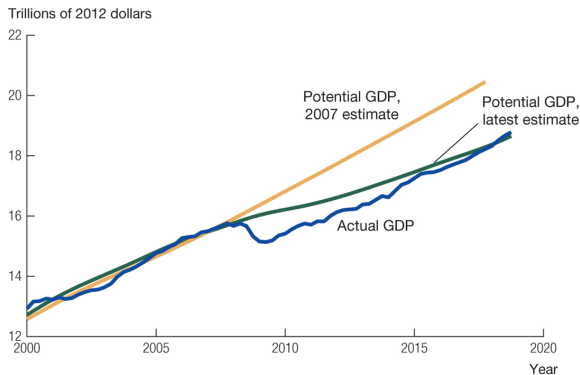
The Fed Funds Rate and the Taylor Rule

- Decline in inflation in 2002 up to the end of 2003
- Fed was concerned about the possibility of deflation at that time
- The decline of inflation was not nearly as sharp as a standard Phillips curve would have predicted during the crisis



How Large is the Output Gap?

- The closing of the gap happened because of revised potential GDP, not from a return to the previous GDP trend
- Why has potential GDP fallen? Low investment rates and TFP growth slowed

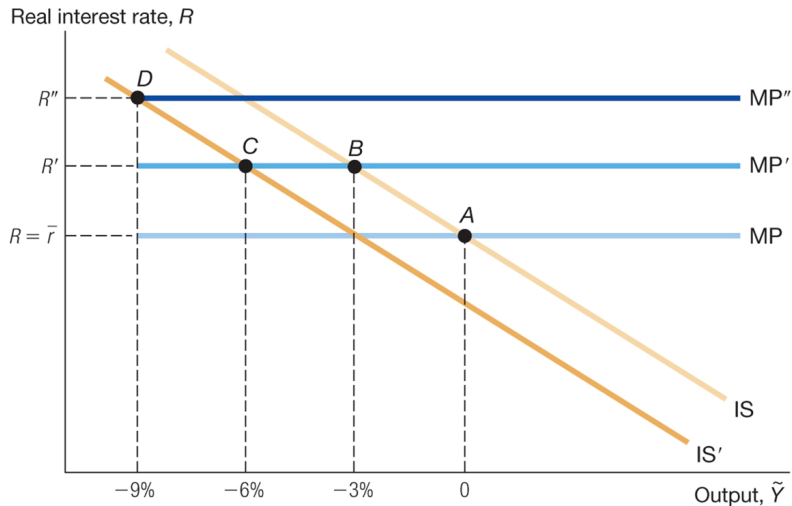


Source: FRED.
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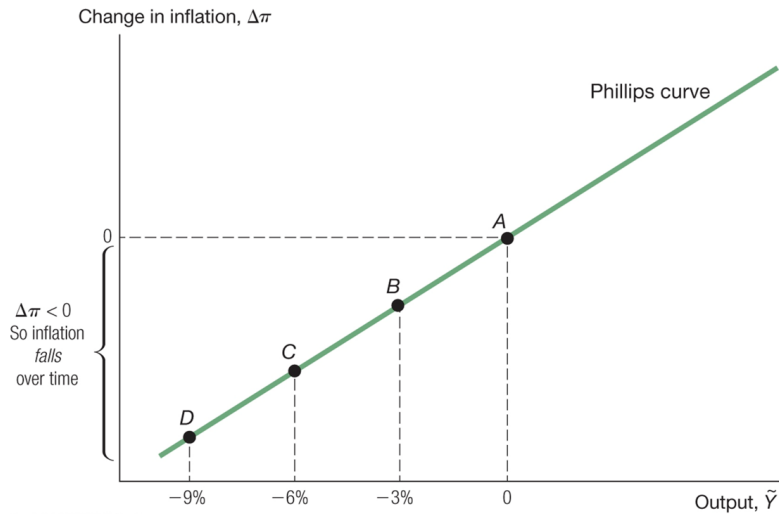
The Money Supply

- Friedman and Schwartz claim that excessively tight monetary policy by the Federal Reserve and the ensuing deflation was the principal cause of the Great Depression
 - They cite that the money supply declined sharply between 1929 and 1933
- Fed is currently focused on stimulating the economy and preventing deflation
 - Expanded the money supply at the end of 2008

The Great Depression in the IS/MP Framework



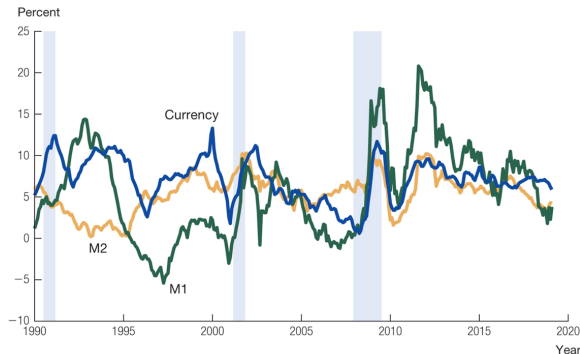
The Phillips Curve and the Great Depression



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The Growth Rate of Various Money Supply Measures

- Currency is the narrowest measure, followed by M1 and M2
- All exhibit rapid growth by the end of 2008
- From the end of 2008 to 2009, the Fed was focused on increasing the money supply, stimulating the economy, and avoiding deflation



Source: The FRED database. Percentage changes over the preceding 12 months.
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The Fed's Balance Sheet

- When conventional monetary policy failed, central bank adopt a set of unconventional monetary policy
 - Allowed these institutions to swap less liquid financial instruments for Treasury securities
 - The provision of liquidity to the large government-sponsored mortgage companies Fannie Mae and Freddie Mac
 - Direct capital injections through the Troubled Asset Relief Program (TARP)

The Fed's Balance Sheet

- These unconventional monetary policy dramatically change the Fed's balance sheet
 - Most assets were U.S. Treasury securities while most liabilities were currency held by the public
- The composition of assets and liabilities also changed
 - On the asset side: the Fed expanded its assets to include lending (through loans or securities) to the rest of the economy, not only to financial institutions but also to nonfinancial corporations
 - On the liability side: additional lending was not financed by printing money. The Fed borrowed from the Treasury and bank excess reserves

The Federal Reserve's Balance Sheet (billions of dollars)

- Various programs are combined into Mortgage-Backed Securities and Other
- The increase are mostly financed by an increase in commercial bank's reserve
 - Fed bought commercial paper or securitized loans from financial institutions and paid for these purchases by crediting their reserve accounts
- But banks kept reserves with the Fed instead of loaning them out
 - Beside that bank were reluctant to make risky loans, the Fed began paying interest on excess reserves – Central bank can adjust this interest rate and control the money flow

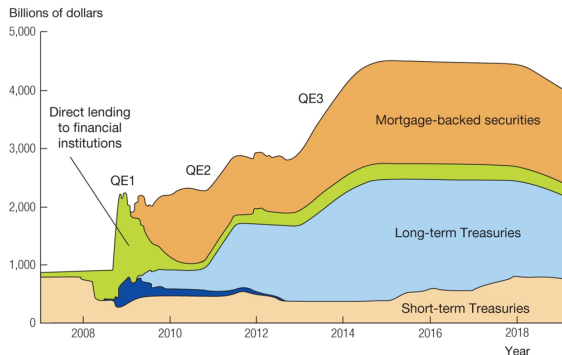
The Federal Reserve's Balance Sheet (billions of dollars)

Assets			Liabilities		
	<i>May 2007</i>	<i>March 2019</i>		<i>May 2007</i>	<i>March 2019</i>
U.S. Treasuries	790	2,180	Currency	814	1,720
Mortgage-backed securities	0	1,590	Reserves	7	1,620
Other	116	230	Other	85	660
<i>Total assets</i>	906	4,000	<i>Total liabilities</i>	906	4,000

Source: Federal Reserve Release H.4.1. See also James Hamilton's "Econbrowser" blog entry "Managing the Fed's Balance Sheet," December 27, 2015.
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Financial Assets of the Federal Reserve

- Purchased a range of securities and made a large volume of loans
- Quantitative easing: central bank purchases of assets other than short-term government bonds
- Initially, the Fed acts as the “lender of last resort”. Then, bought only MBS in QE1 and bought long-term U.S. Treasuries in QE2. During QE3, they bought both



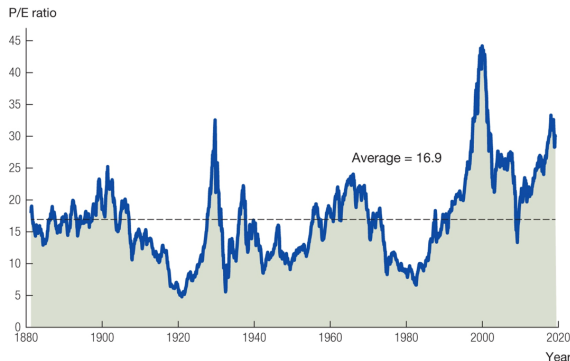
Source: Credit Easing, Federal Reserve Bank of Cleveland. Accessed on 09/27/2019. <https://www.clevelandfed.org/en/our-research/indicators-and-data/credit-easing.aspx>. Reproduced with permission.

Should Monetary Policy Respond to Asset Prices?

- It is now evident that there was a housing bubble in the mid-2000s
- Should the Fed attempt to curtail asset price bubbles when they occur? The Fed Chairman at that time Bernanke argued in 2000
 - Bubbles are difficult to identify in real time
 - Monetary policy is too imprecise to deal with bubbles
 - Policymakers should use more refined instruments:
 - capital requirements
 - regulation of lending standards

The Price–Earnings Ratio in the Stock Market

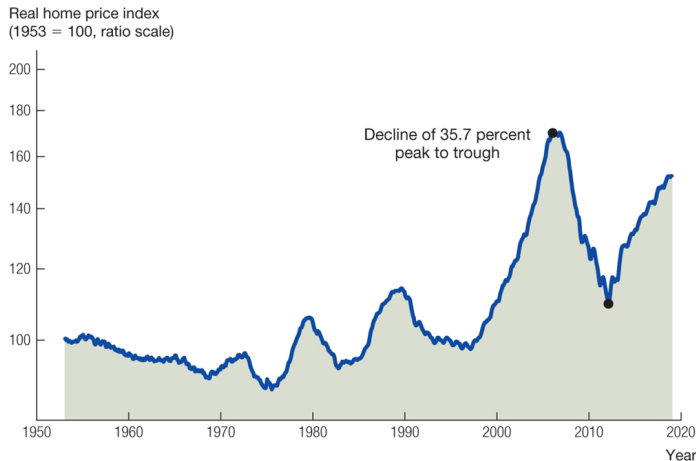
- Often serve as a measure of bubbles
- In asset pricing models, stock prices should be equal to the present discounted value of future profits
- Whenever the ratio of stock prices to company earnings gets too far away from its mean, it tends to revert back



Source: Robert Shiller, www.econ.yale.edu/~shiller/data.htm.
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Real Home Prices in the United States

- By nearly any measure, the appreciation of home prices between 2000 and 2006 was unusual
- Moreover, previous episodes of rapidly rising real home prices were often followed by real declines



The Troubled Asset Relief Program

- Economists agree on the importance of restoring the financial system, but debate surrounds the most effective policy
- Policies such as the Troubled Asset Relief Program (TARP) have been implemented by the Federal Reserve and Department of the Treasury

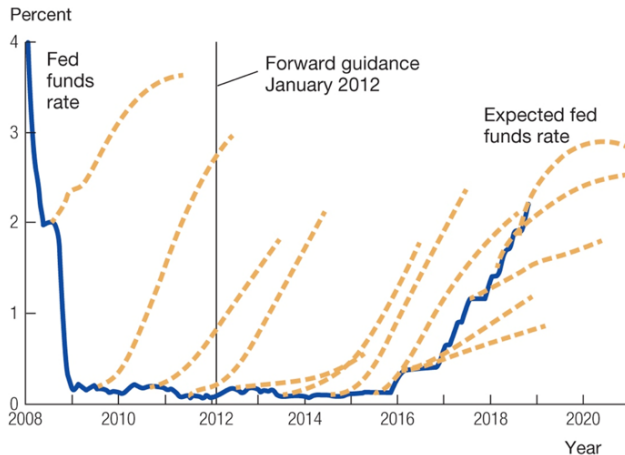
The Troubled Asset Relief Program

- TARP included measures like purchasing toxic assets and injecting capital into financial institutions
 - Purchases of “toxic” assets: Banks possess bad assets, which limits lending
 - Capital injections into financial institutions: Original TARP invested \$25 billion in each large financial institution
- In the fall of 2008, TARP established a \$700 billion fund for the Treasury to purchase toxic assets
 - purchase equity in banks and other financial institutions, to guarantee loans to those institutions, and even to bail out some U.S. automakers
- Complete reorganization of financial institutions: reorganize debt into new equity claims for the former debt holders
- The program did not actually incur large losses; as of the end of 2012, about 97 percent of the disbursements had been returned to the Treasury

- American Recovery and Reinvestment Act (ARRA) 2009
 - Tax cuts and new government spending – More than \$250 billion in tax cuts and More than \$500 billion in new government spending
 - Increased the deficit to 10 percent of GDP in 2009 (was 3% of the GDP in 2008)
- Most economists believed there should be a fiscal stimulus
- Disagreement stemmed from the type of spending and weight of tax cuts
 - The Ricardian equivalence argument was also a factor
 - High spending must be financed by higher taxes in the future
 - The prospect of these taxes may reduce the current impact of the stimulus package
 - Temporary measures that seek to change the timing of production may overcome these concerns (e.g., investment tax credit, etc)

Forward Guidance

- The dashed lines are the market expectations about the future fed funds rate
- The market seems to believe the interest rate would stay low for sometime after Jan 2012



Source: Reprinted from the Federal Reserve Bank of San Francisco's Economic Letter, "A Review of the Fed's Unconventional Monetary Policy," vol. 2018-27, December 3, 2018, by Glenn D. Rudebusch. The opinions expressed in this article do not necessarily reflect the views of the management of the Federal Reserve Bank of San Francisco or

The European Debt Crisis

- Financial crisis was not a U.S. phenomenon
- Sovereign debt crisis in Europe
 - Government debt of countries like Greece, Ireland, Italy, and Spain came under pressure
 - Interest rates on the debt rising as high as 7 percent
 - Severe problems in the banking sector
 - High unemployment

- A key downside to the way in which the financial crisis was handled by policymakers is that it has made problems of moral hazard more severe
 - Many of these institutions were deemed too systemically important to fail or too big to fail
 - Bankers would expect the banking sector will be bailed and would take any risk
- How do we prevent major problems in the financial system?
 - Gain greater understanding of volatile prices (e.g., housing, stocks, bubbles)
 - Acknowledge costs with the benefits of financial intervention
 - Understand the downside of moral hazard – minimize incentive for bankers to take “heads, I win; tails, the economy loses” types of bets

- How do firms fail under normal circumstances?
 - A common solution when a firm's liabilities surpass its assets is to declare bankruptcy
 - A judge or some other government entity steps in and reorganizes the firm if possible so that it can resume its business
 - Equity (net worth) is already zero or even negative, so the stockholders have lost their entire investment
 - Debt is written down to zero, and the former debtholders are given equity claims in the newly reorganized firm
 - The value of the debt at the time of the reorganization becomes the new equity in the new firm

- This approach has a number of appealing features for financial sector as well
 - Stockholders and bondholders bear the brunt of the burden, not taxpayers
 - With the bank recapitalized, it should emerge with the ability to lend
- Potential problem with this approach
 - Severely interferes with the functioning of financial markets if not handled carefully
 - Liquidity problem in the short run
 - The panic following the collapse of Lehman Brothers in September 2008 is a reminder of what can go wrong

Squam Lake Group suggested guidelines for financial reform:

- Create a systemic regulator
 - Each country should have one primary regulator charged with monitoring systemic risk in financial markets
 - Responsible for enforcing the other elements of financial reform
- Enhance capital requirements
 - The minimum percentage of capital requirements (equity/capital to asset ratio) should be raised so that financial institutions have a larger “cushion” in case the value of their assets declines suddenly
 - Also, larger or more complex financial institutions should face higher capital requirements
- Main cost of bailouts
 - Borne by people outside of the finance industry
 - Measured by lost jobs and forgone GDP

- Link executive compensation to long-term performance
 - The compensation of senior management in systemically important financial institutions should be tied, at least in part, to the long-term performance of the firms
- Require “living wills.”
 - Some important financial institutions will inevitably fail
 - A living will—a set of instructions for how this failure should be carried out to keep to a low level the disruption to financial markets—would minimize the effects of such a failure

- Require convertible debt
 - Systemically important financial institutions might also be required to issue a significant amount of a new, hybrid debt instrument that converts to equity when certain prespecified crisis conditions arise
 - This hybrid is called convertible debt
 - When the value of equity in such a firm falls significantly, this debt would convert to equity, recapitalizing banks at no cost to taxpayers
 - Bondholders and shareholders would themselves bear the costs of financial problems, minimizing the likelihood of taxpayer bailouts
 - This convertible debt functions like an automated, limited version of the reorganization that occurs in bankruptcy, but it should have lesser effect on the financial markets than a more formal reorganization would

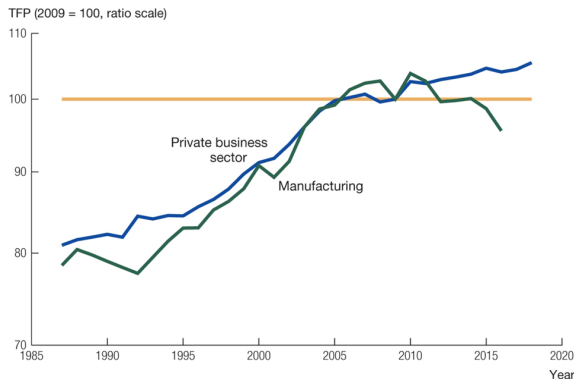
The Aftermath of the Great Recession

The slow recovery following the Great Recession has prompted two proposed explanations:

- Secular stagnation, characterized by sluggish economic growth, persistently low inflation, and negative real interest rates
 - Japan has experienced with this phenomenon since the 1990s
- The combination of secular stagnation and a "global saving glut" has depressed investment demand while simultaneously increasing the supply of saving, resulting in negative real interest rates
- The presence of the zero lower bound does not allow the market to be cleared
- Reduced investment leads to less capital accumulation. As indicated in the Solow model, the economic growth is slower

The Aftermath of the Great Recession

- TFP growth has been slow since 2004
- R& D spending remains strong. But, it may be because the financial crisis increased the misallocation of resources



Source: Bureau of Labor Statistics multifactor productivity database.
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Macroeconomic Research after the Financial Crisis

- Did macroeconomists fail?
 - Research was not available to provide guidelines
 - Research has been limited by computational power and data availability
- New models are more advanced and include
 - Frictions: sticky inflation and limited financial markets
 - Heterogeneous consumers – some are rich, some are poor; some are more willing to save than the other, etc
 - Imperfect financial markets – agents cannot buy any asset or combination of asset such that it would cancel out all negative shock
 - Allowing for more roles of government
- Economist response with a sharp increase in macroeconomic research focused on economic fluctuations and financial imperfections

Conclusion

- The global financial crisis will likely go down in history as the largest recession since the Great Depression
- The “Great Recession” was a balance sheet crisis for
 - Financial institutions
 - Households
 - The Federal Reserve
 - The government
- As we recover from the crisis, the question remain, is the productivity slow down temporary?
 - The experience from Great Depression would suggest no...

Next Lecture

- Next lecture will cover Ch.15